

OCT Retinal Disease Classification Using Graph Neural Networks

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Submitted To

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Focus Area

Multi-class retinal disease classification from Optical Coherence Tomography (OCT) images using Graph Neural Networks, with explainability via SHAP and LIME.

Research Objectives

This study establishes a complete end-to-end pipeline for retinal disease classification, benchmarking classical machine learning approaches against a novel Graph Neural Network architecture designed to preserve spatial retinal structure.



Build a Complete Pipeline

Construct a full retinal disease classification pipeline using the OCTDL dataset, from raw OCT images through feature extraction to final prediction.



Preserve Spatial Information

Improve classification performance by retaining the spatial structure of retinal tissue, which is lost in conventional feature-vector approaches.



Benchmark Baselines vs. GNN

Compare multiple machine learning baselines — including SVM, XGBoost, LightGBM, MLP, and 1D CNN — against the proposed Graph Neural Network architecture.



Explain Model Decisions

Apply Explainable AI techniques — specifically SHAP and LIME — to interpret and validate model predictions in a clinically meaningful way.

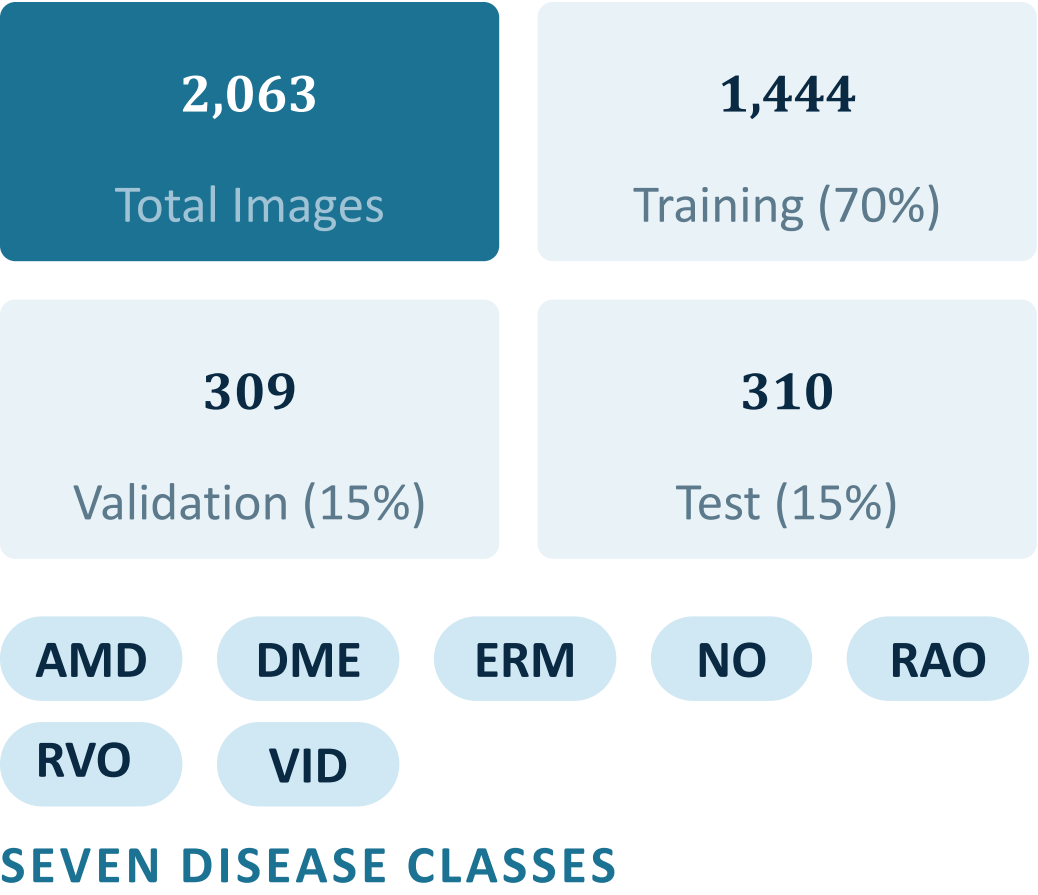
DATASET

Dataset Information

Dataset	OCTDL — Optical Coherence Tomography Dataset
Source	Kaggle
Problem Type	Multi-class Image Classification
Classes (7)	AMD, DME, ERM, NO, RAO, RVO, VID

Why Subject-Level Split?

- ✓ One patient's images never appear in multiple sets.
- ✓ Prevents data leakage across train / val / test.
- ✓ Ensures a fair, unbiased evaluation.



Imbalance note NO and AMD have the most samples; RAO and RVO are the most under-represented classes.

Exploratory Data Analysis & Research Gap

EDA SCOPE

- Cleaning & corrupted-image removal
- Class distribution analysis
- Sample visualization
- Duplicate checking
- Image resolution (width/height) stats
- Train / val / test verification

Key Findings

- Seven retinal disease categories in the dataset.
- Images are RGB OCT scans.
- Dataset is imbalanced overall.
- NO and AMD have more samples.
- RAO and RVO contain the fewest samples.

RELATED WORK (2022–2026)

VGG16	98.7% Accuracy
Graph-Fused CNN + ViT	99.8% Accuracy
GCN + U-Net	Dice 0.850
Explainable Graph Model	61.8% ISNT alignment
Hybrid CNN-ViT	Better than pure ViT

Research Gap

- Loss of spatial retinal information
- Naive feature fusion
- Poor robustness across imaging devices
- Limited integration of clinical metadata

Task 2: Baseline Models

Preprocessing Pipeline

1

Subject-Level Split

Patient IDs kept separate across all folds to prevent leakage.

2

ResNet-50 Feature Extraction

Deep features extracted from pretrained ResNet-50 as input representations.

3

Scaling & Feature Selection

StandardScaler normalization, Variance Threshold filtering, and Mutual Information selection applied.

4

Class Balancing

Class-weight balancing applied to all models; SMOTE oversampling applied specifically for k-NN.

Models Tested

Linear & Kernel

- Logistic Regression
- SVM (RBF kernel)
- k-Nearest Neighbors

Tree-Based

- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM

Neural

- MLP (Multi-Layer Perceptron)
- 1D CNN

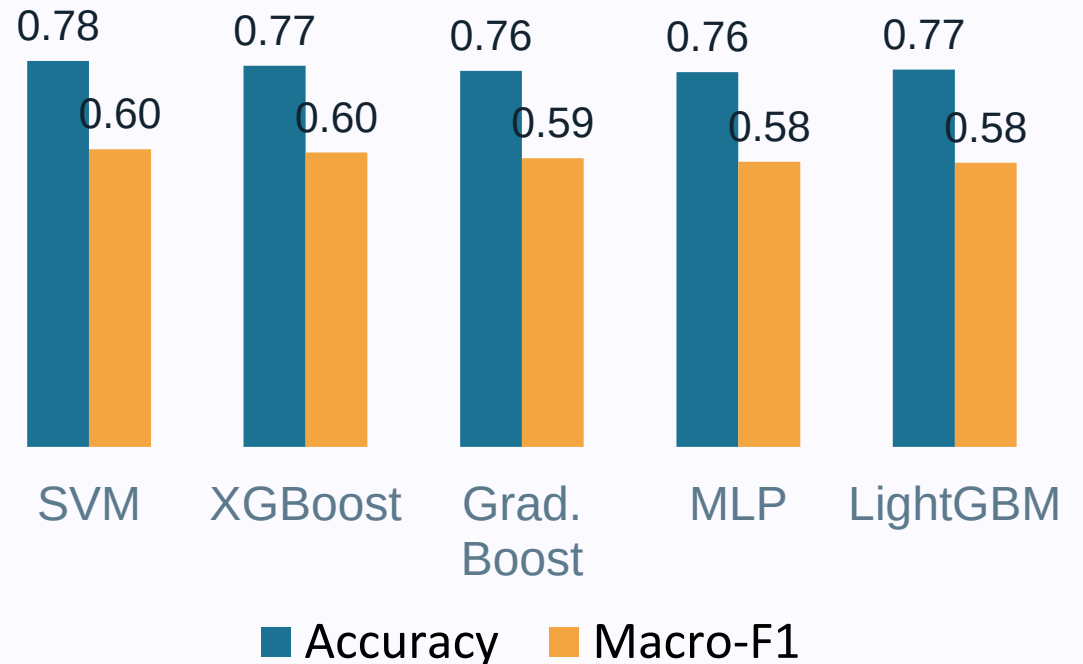
Macro-F1 was chosen as the primary metric to give equal importance to rare disease classes regardless of class frequency.

Task 2-Baseline Performance Results

Top-performing baselines ranked by Macro-F1. **SVM (RBF)** emerged as the strongest baseline and the model to beat for the proposed GNN.

BASELINE PERFORMANCE (TOP 5)

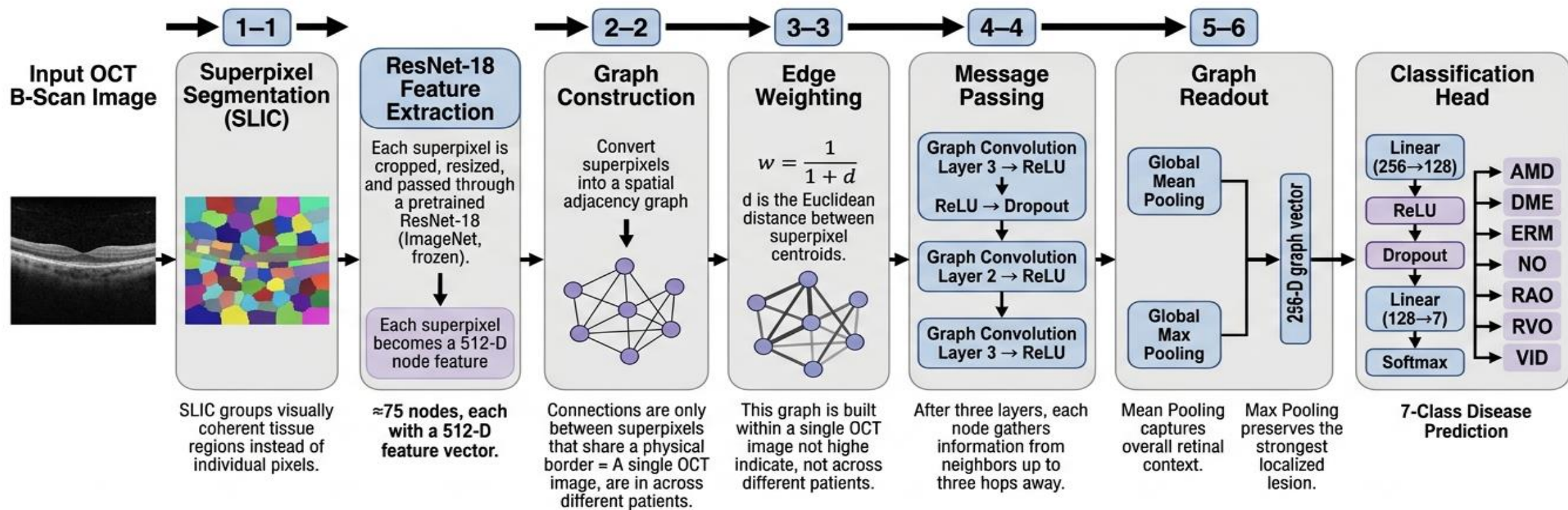
Model	Accuracy	Macro-F1	ROC-AUC	Time (s)
SVM ☆	0.7845	0.6049	0.9151	4.10
XGBoost	0.7744	0.5981	0.8807	6.75
Gradient Boosting	0.7644	0.5866	0.8567	180.46
LightGBM	0.7669	0.5772	0.8861	4.76
MLP	0.7619	0.5794	0.8803	1.15



Model to Beat: SVM achieved the highest Macro-F1 (0.6049), making it the strongest baseline. Macro-F1 was chosen as the primary metric because it gives equal importance to rare classes. SVM (RBF) was the strongest classical baseline. The proposed GNN was designed to outperform it.

TASK 2 · Architecture of our work

Proposed GNN Architecture for OCTDL Retinal Disease Classification



Training Protocol



Loss Function:
Weighted Cross-Entropy using inverse class frequency



Optimizer:
Adam



Early Stopping:
Monitor validation Macro-F1 and stop if performance no longer improves.



Best Checkpoint:
Save the model with the highest validation Macro-F1 instead of the last epoch.



Purpose:
Prevent rare classes from being ignored and improve generalization.

Why Graph Neural Networks?

Limitations of Baseline Approaches

Spatial Information Lost

Every OCT image is flattened into a single feature vector. The spatial location of retinal lesions — critical for distinguishing AMD from DME — is entirely discarded after global pooling.

Indistinguishable Lesion Positions

Two lesions in different retinal regions produce similar representations after pooling, making it impossible for the model to reason about where pathology occurs.

1D-CNN Structural Blindness

The 1D-CNN captures only a single scan line at a time, missing the full 2D retinal structure that defines disease morphology.

GNN Architectural Advantages

1. Superpixel Graph Construction

Each OCT image is converted into a graph of ~75 superpixels, preserving the spatial arrangement of retinal tissue regions.

2. Spatial Adjacency Edges

Neighboring tissue regions are connected using spatial adjacency, encoding the anatomical relationships between retinal layers.

3. Message Passing

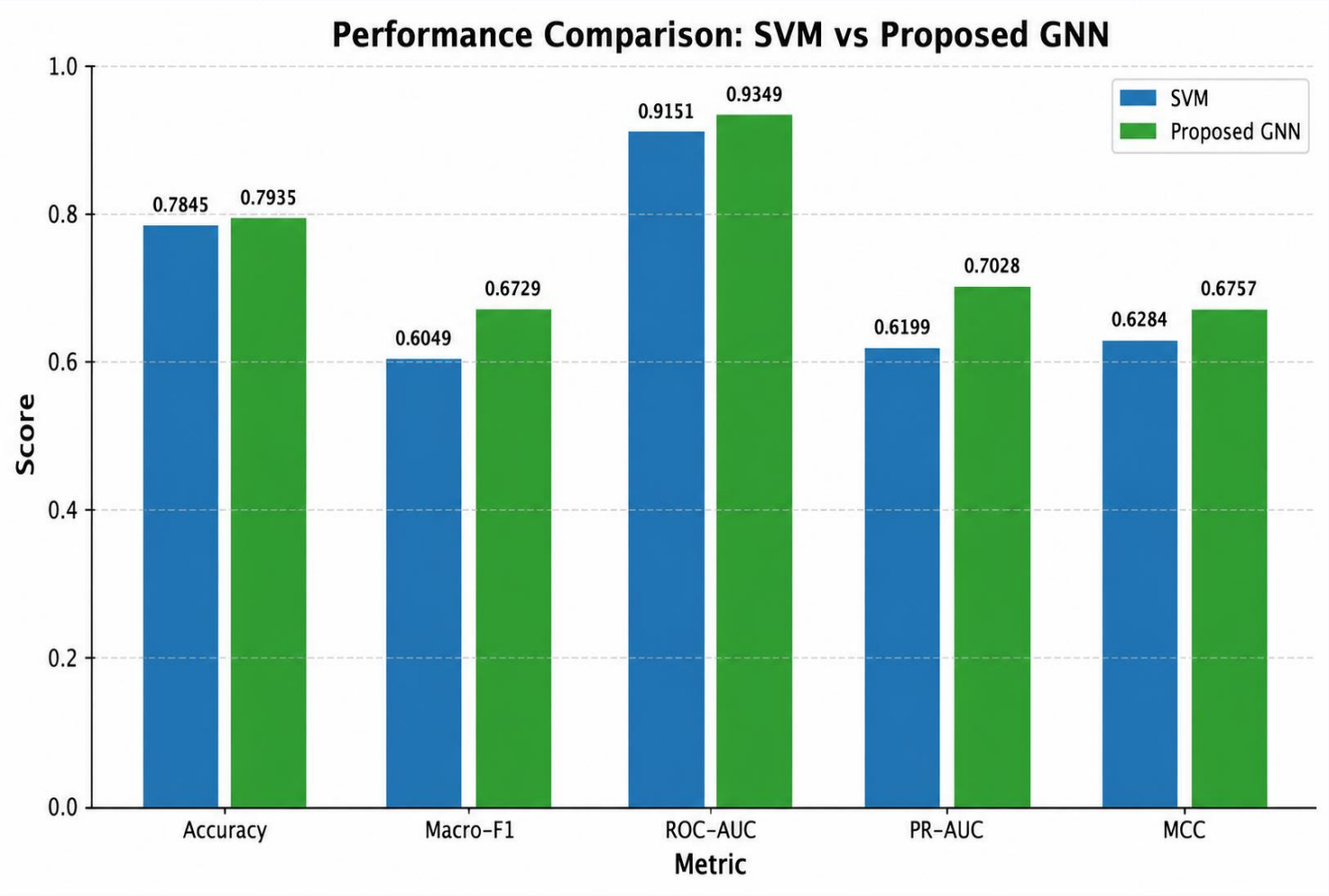
Each region learns from nearby healthy and diseased tissues through iterative message passing, enabling context-aware classification.

4. Mean + Max Pooling

Dual pooling captures both overall tissue context (mean) and localized lesion signals (max), improving detection of AMD, DME, and ERM.

Task-2 RESULTS

GNN Results Vs Baseline Performance Comparison



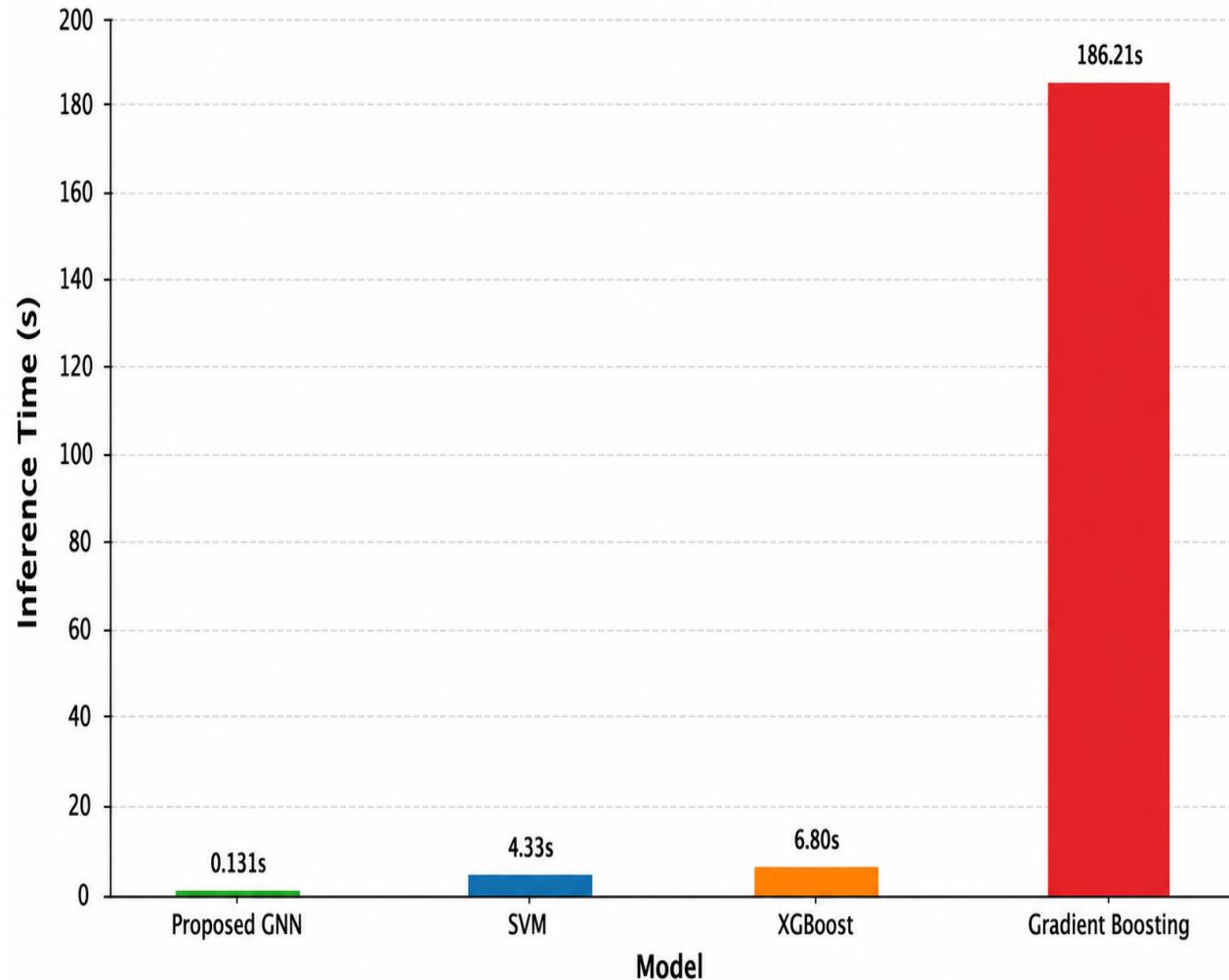
IMPROVEMENT OVER BEST BASELINE (SVM)

Metric	SVM	Proposed GNN
Accuracy	0.7845	0.7935
Macro-F1	0.6049	0.6729
ROC-AUC	0.9151	0.9349
PR-AUC	0.6199	0.7028
MCC	0.6284	0.6757

Task-2 RESULTS

GNN Results Vs Baseline Performance Comparison

Inference Time Comparison



Why Did It Improve?

- 1 Preserved Spatial Structure**
Retinal layout is retained through graph message passing rather than flattened.
- 2 Region-Level Reasoning**
Predictions draw on localized superpixel regions, not just global features.
- 3 Neighborhood Information**
Message passing lets nearby regions inform each other's representation.
- 4 Localized Lesion Handling**
Better sensitivity to small, spatially concentrated pathological patterns.

TASK 3 · MODEL DEVELOPMENT

Ablation Study & Model Improvement

Starting from the initial GCN (A0), we changed one or two components at a time under a fixed dataset, patient-level split, and early-stopping protocol — using validation Macro-F1 for selection while keeping the test set untouched.

Run	Main Change	Val F1	Test Macro-F1
A0	Initial GCN	0.506	0.386
A1	Larger GCN + Regularization	0.685	0.539
A2	Distance-weighted edges	0.684	0.608
A3	Global Fusion (Final Model)	0.712	0.531
A4	k-NN Graph	0.588	0.511
A5	GAT	0.658	0.579
A6	GraphSAGE	0.572	0.502
A7	Adam Optimizer	0.669	0.583
A8	No LR Scheduler	0.400	0.400

KEY FINDINGS

- 64→128 hidden units + 2→3 GCN layers gave the largest lift (0.386 → 0.539).
- Distance-weighted edges pushed performance further (0.539 → 0.608).
- Spatial adjacency beat k-NN graph construction.
- GCN outperformed both GAT and GraphSAGE on this dataset.
- AdamW + ReduceLROnPlateau was most stable; removing the scheduler caused the biggest drop (0.712 → 0.400).

Final Selection: A3 chosen for highest validation Macro-F1 (0.712).

TASK 3 · MODEL DEVELOPMENT

Final Performance: Fair, Leakage-Free Comparison

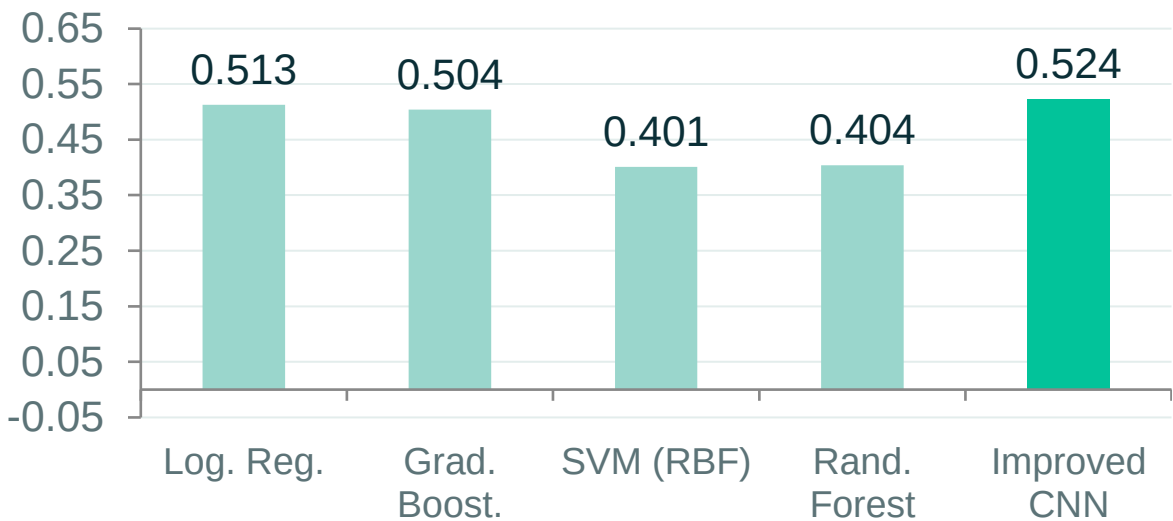
EVALUATION PROTOCOL

- Same patient-grouped split for every model.
- Test set untouched until the final model was selected.
- Classical ML: ResNet-50 (ImageNet) features → StandardScaler → SMOTE → classifier.
- Improved CNN trained end-to-end under the same protocol.

Best Baseline
The Improved CNN became the strongest classical baseline.

Model	Accuracy	Macro-F1
Logistic Regression	0.774	0.513
Gradient Boosting	0.771	0.504
SVM (RBF)	0.742	0.401
Random Forest	0.720	0.404
Improved CNN ☆	0.787	0.524

Macro-F1 by Model



TASK 3 · MODEL DEVELOPMENT

Final Model Comparison

Model	Accuracy	Macro-F1
Best Baseline (Improved CNN)	0.787	0.524
First GNN (A0)	0.697	0.281
Final GNN (A3) ☆	0.803	0.531

0.281 → 0.531

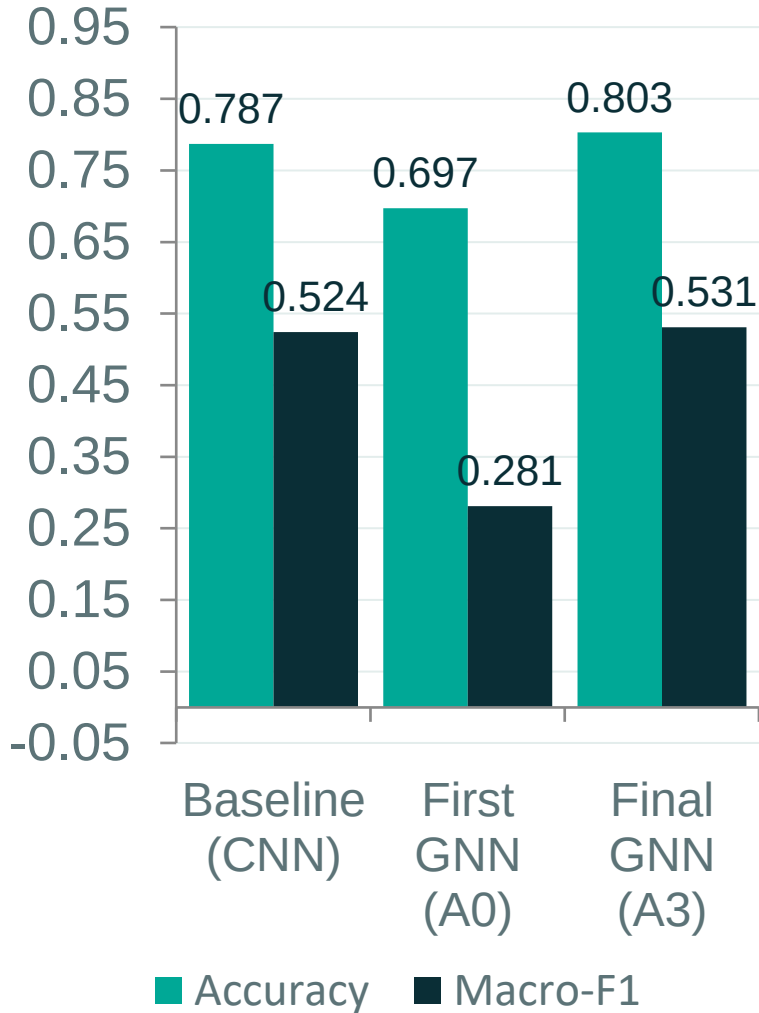
Macro-F1 improvement,
first GNN → Final GNN

+0.007

Macro-F1 vs. best
baseline (Improved CNN)

The optimized GNN improved Macro-F1 from 0.281 to 0.531 vs. the first GNN, and edged out the strongest baseline — while preserving a fair, leakage-free evaluation.

Accuracy & Macro-F1



Cross-Validation & Significance Testing

WHY CROSS-VALIDATION?

A single train/test split can favor one model by chance. We used 5-fold StratifiedGroupKFold to check whether the Final GNN consistently outperformed the best baseline — with patient IDs kept separate across folds to prevent leakage.

Fold 1

Fold 2

Fold 3

Fold 4

Fold 5

Patients grouped and rotated across 5 folds — never split within a patient.

CROSS-VALIDATION SETUP

- 5 patient-grouped folds (1,749 images / 697 patients for train + validation).
- Both models retrained from scratch on every fold.
- Class-weighted loss across all folds.
- AdamW optimizer with learning-rate scheduling.
- Early stopping on validation Macro-F1.

Mean Performance Across 5 Folds

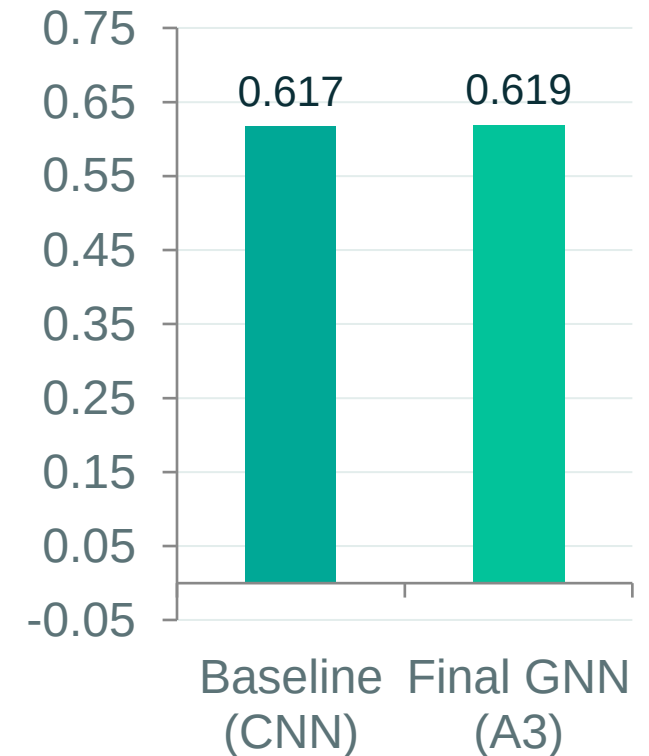
Model	Accuracy	Balanced Acc.	Macro-F1
Best Baseline (Improved CNN)	0.786 ± 0.052	0.677 ± 0.085	0.617 ± 0.104
Final GNN (A3)	0.771 ± 0.068	0.624 ± 0.044	0.619 ± 0.066

+0.002

Mean Macro-F1 advantage (GNN vs. baseline)

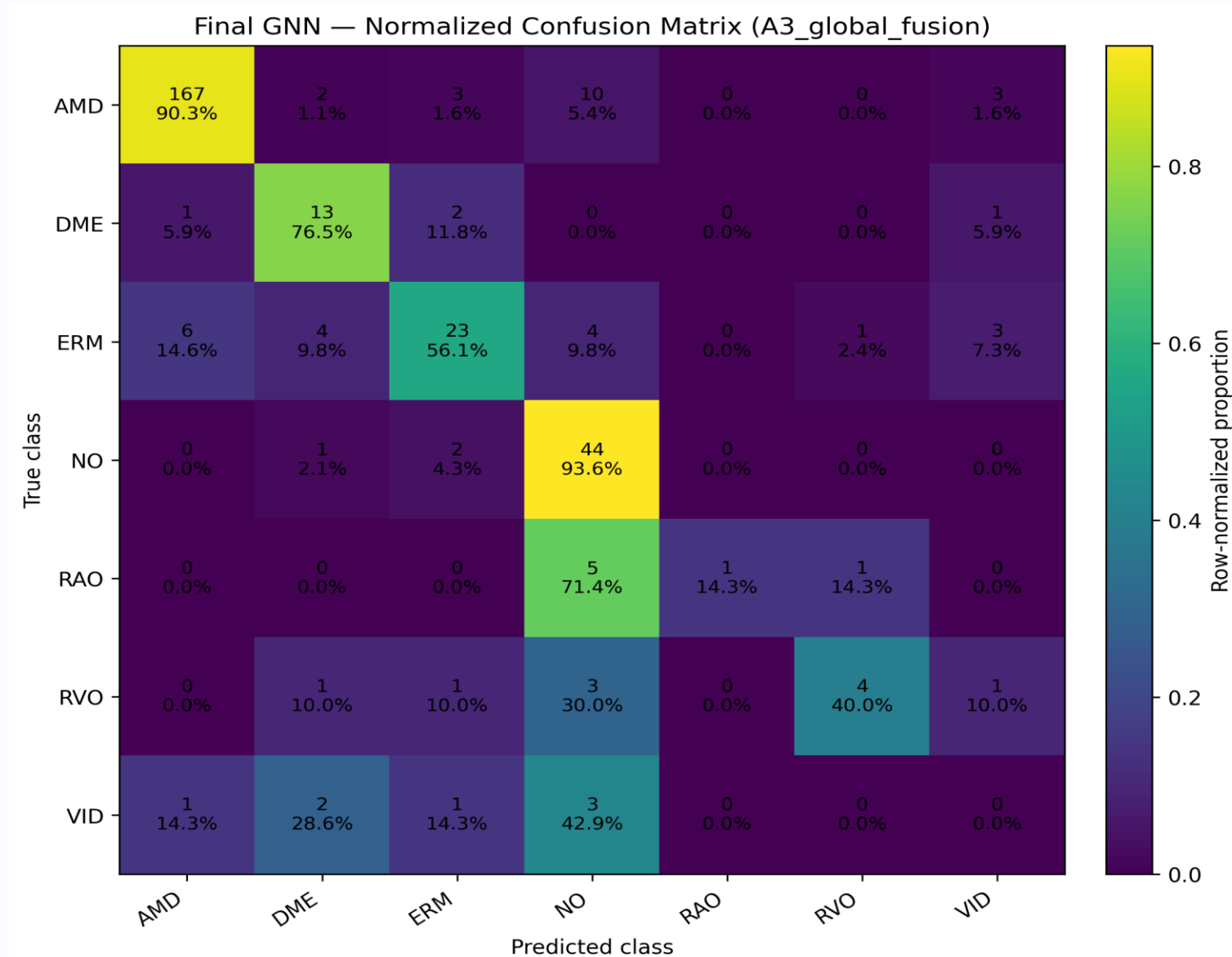
⚠️ Fold-to-fold variation (± 0.066 – 0.104) is much larger than the $+0.002$ gap between models.

Macro-F1 (Mean $\pm$ SD)



The Final GNN reached the highest average Macro-F1 (0.619) — a marginal edge over the baseline's 0.617 — but the gap is small next to natural fold variation, so a formal significance test is needed before claiming an advantage.

Final GNN Confusion matrix



Statistical Significance Testing

We tested whether the Final GNN's improvement over the Improved CNN baseline was real or due to chance, using two tests for two evaluation settings.

PRIMARY · Wilcoxon Signed-Rank (5-Fold CV)

Statistic	7.0
p-value	1.0000
Significance Level (α)	0.05

Result: Not Significant

Mean Macro-F1 Final GNN: 0.6194 · Improved CNN: 0.6175

SECONDARY · Exact McNemar (Held-out Test Set)

Baseline-only Correct	12
GNN-only Correct	17
Discordant Pairs	29
Statistic	12.0
p-value	0.4583

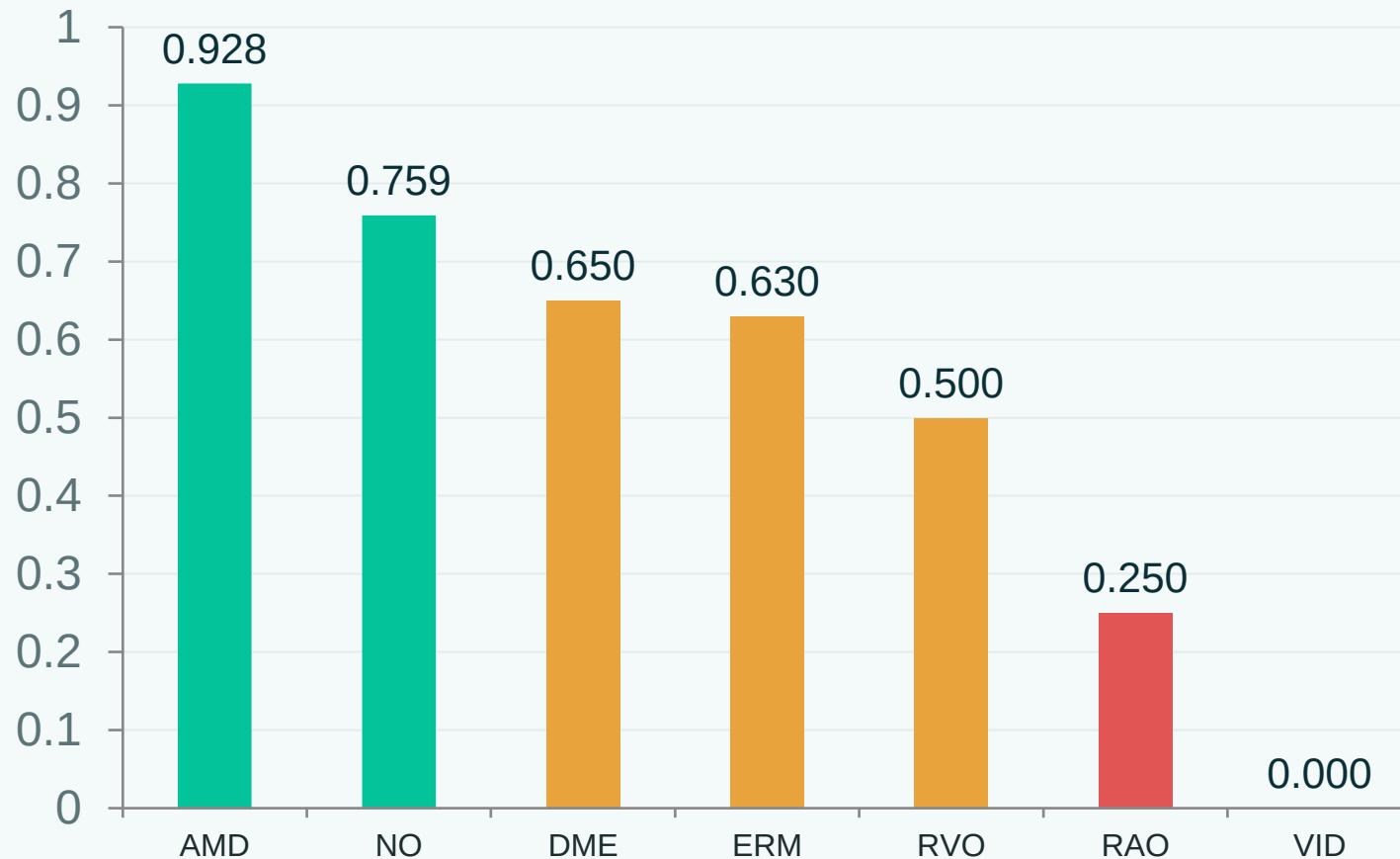
Result: Not Significant

McNemar used only for the shared held-out test set.

Both tests reach the same conclusion: although the Final GNN scored slightly higher, the improvement is not statistically significant at $\alpha = 0.05$.

Final Model Diagnostics

Per-Class Performance — Final GNN



KEY FINDINGS

- AMD achieved the best performance (F1 = 0.928).
- NO was also classified accurately (F1 = 0.759).
- Rare classes — RAO, RVO, VID — showed weaker performance due to limited training samples.
- VID was the most challenging class, with no correct predictions on the test set.

Class imbalance remains the model's core limitation for rare retinal conditions.

Explainability (SHAP + LIME)

LIME and SHAP were used to understand why the Final GNN made its predictions — explanations were generated for the highest-confidence correct prediction (RAO) and the highest-confidence wrong

LIME

- Highlights the important superpixels that influenced each prediction.
- Correct RAO prediction: focus centered on abnormal retinal regions.
- Wrong RVO prediction: overemphasized normal-looking tissue.

SHAP

- Shows each region's positive and negative contribution to the prediction.
- Confirmed the same important regions identified by LIME.
- Revealed the misclassified RVO image received weak evidence for its true class.

Key Takeaway: SHAP and LIME made the Final GNN's decisions more interpretable by showing which retinal regions influenced both correct and incorrect predictions.

LOOKING AHEAD

Future Work

1 Rare-Class Recognition

Focus on RAO, RVO, and VID, which have limited test samples and drive performance variability.

2 Broader Explainability

Evaluate LIME and SHAP across many samples from all disease classes, not just one correct/incorrect case.

3 Generalization

Validate the Final GNN on external datasets and different imaging devices / settings.

4 Graph Design

Explore alternative graph-construction and GNN strategies to improve spatial representation.

5 Robustness to Imbalance

Investigate methods that reduce sensitivity to class imbalance for underrepresented diseases.

6 Broader Validation

Use larger, more diverse datasets for more reliable estimates of model performance.

These directions follow directly from the report's limitations and discussion — particularly its emphasis on rare classes, broader explainability, and external / device generalization.

Conclusion

80.3%

Test accuracy — Final GNN (A3)

0.531

Test Macro-F1, vs. 0.524 baseline

0.619 vs 0.617

5-fold CV Macro-F1 — not statistically significant

A Competitive, Interpretable Approach

- Built a patient-grouped, leakage-free evaluation pipeline for multi-class OCT disease classification.
- The Final GNN (A3) performed competitively against the Improved CNN baseline.
- Ablation analysis showed the value of regularization, distance-weighted spatial edges, global-image fusion, AdamW, and LR scheduling.
- LIME and SHAP provided interpretable visual explanations for both correct and incorrect predictions.
- The main remaining challenge is reliable classification of rare disease classes — RAO, RVO, and VID.

The proposed GNN is best characterized as a competitive and interpretable approach — rather than a statistically proven improvement over the baseline.

Thank You